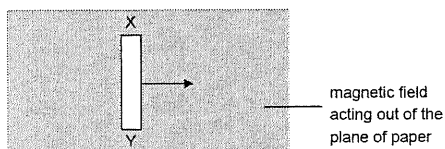


Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [AJC/H2/Prelims 2010/P1/31] (Amended)

A metal rod XY is moving to the right, perpendicular to a magnetic field as shown.



Which row correctly describes the magnetic force acting on the rod and the potential of the rod XY?

	magnetic force on rod	potential of rod
A	directed to the left	Y is higher than X
B	directed to the right	Y is higher than X
C	no force	Y is higher than X
D	no force	X is higher than Y

2. [AJC/H2/Prelims 2010/P1/32]

A solid copper plate is oscillating between two magnets as shown in Fig. 2.1. The solid plate comes to a rest shortly. The solid plate is now changed to another plate with slits of the same material as shown in Fig. 2.2. The new plate takes a slightly longer time before it comes to a rest.

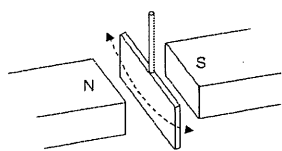


Fig. 2.1

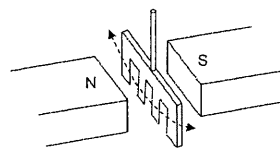


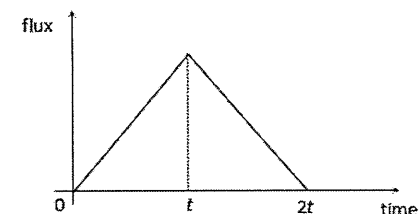
Fig. 2.2

What is the correct reason for this effect?

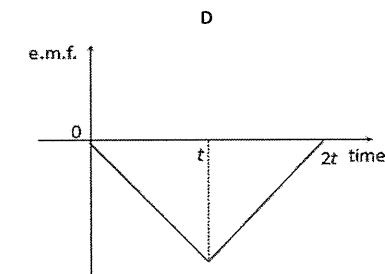
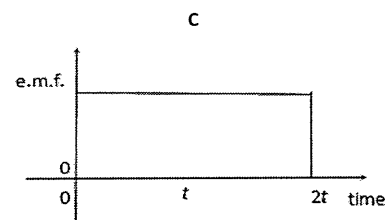
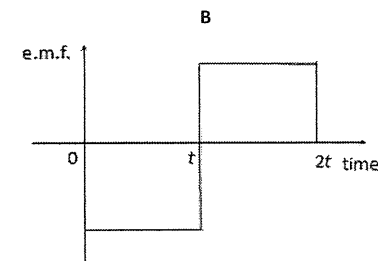
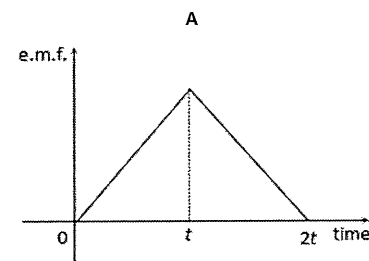
- A The amount of eddy current flowing round the plate is reduced.
- B The amount of ferromagnetic material is reduced.
- C The plate is less magnetized.
- D The mass of the plate is reduced.

3. [CJC/H2/Prelims 2010/P1/33]

The graph shows the variation with time of the magnetic flux linking a coil.

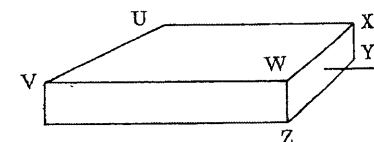


Which graph shows the variation with time of the e.m.f. induced in the coil?



4. [Dunman High/H2/Prelims 2010/P1/30]

A rectangular conductor is moving horizontally from left to right in a uniform magnetic field perpendicular to UVWX.

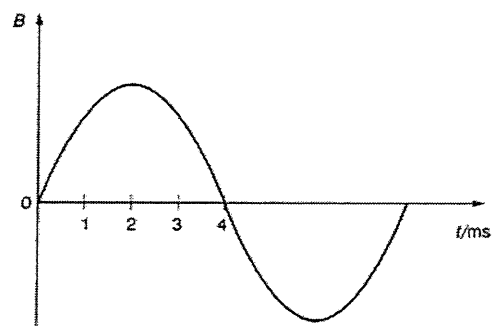


Which of the following pairs of points will NOT be at the same potential?

- A U and V
- B U and X
- C V and W
- D W and Z

5. [HCI/H2/Prelims 2010/P1/30]

An e.m.f. is induced in a coil placed in a changing magnetic field. The flux density B of this field varies with time t as shown.

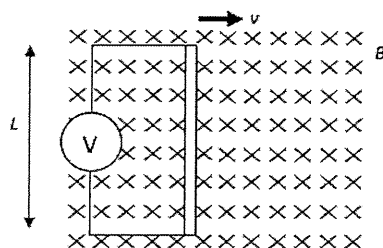


At which value of t is the magnitude of the e.m.f. induced in the wire a maximum?

- A 1 ms B 2 ms C 3 ms D 4 ms

6. [HCI/H2/Prelims 2010/P1/31]

A copper bar of length L is moving to the right with a uniform speed v in a region of uniform magnetic field of flux density B , directed perpendicularly downwards into the paper as shown.



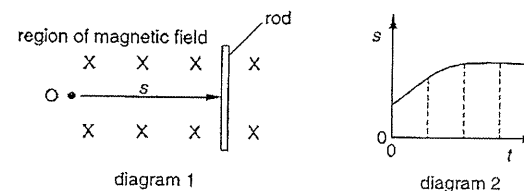
The ends of the rods are rigidly connected to a voltmeter which moves with the rod.

What is the reading on the voltmeter?

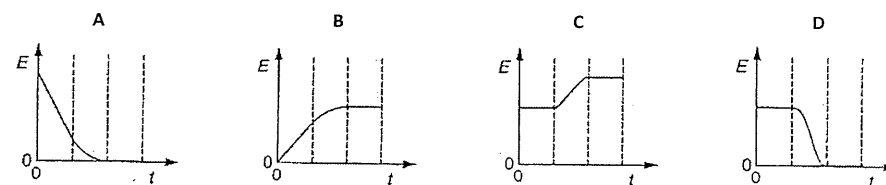
- A zero
B non-zero reading less than BLv
C BLv
D more than BLv

7. [UC/H2/Prelims 2010/P1/31]

An aluminium rod moves at right angles to a uniform magnetic field as shown in diagram 1. Diagram 2 shows the variation with time t of the distance s from point O.

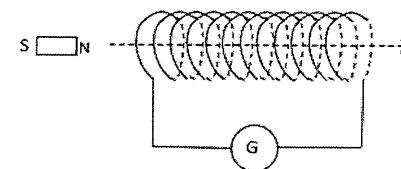


Which graph best shows the variation with t of the e.m.f. E induced in the rod?

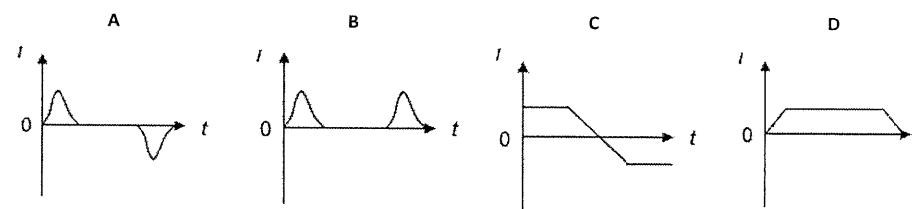


8. [MJC/H2/Prelims 2010/P1/31]

A short bar magnet passes at a steady speed through a long solenoid. A galvanometer is connected across the solenoid.

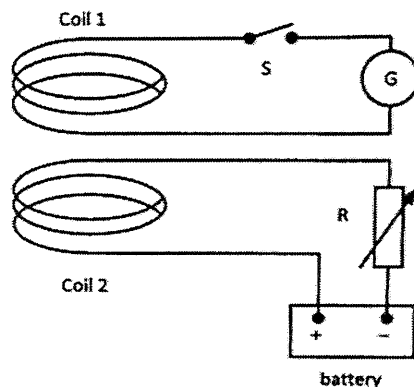


Which graph best represents the variation of the current I of the galvanometer with time t ?



9. [MI/H2/Prelims 2010/P1/32]

Two flat horizontal coils are mounted as shown.



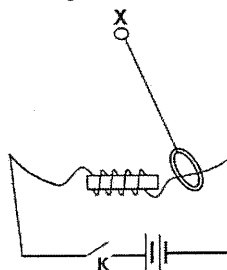
Which of the following actions will cause the galvanometer to register a current?

1. With switch S closed, Coil 2 is stationary while Coil 1 is moved upwards.
2. Both coils are stationary and S is switched on and off.
3. With switch S closed, the variable resistor R is increased and decreased rapidly.

A 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

10. [Dunman High/H2/Prelims 2010/P1] (Amended)

A copper ring is suspended by a long, light rod pivoted at X so that it may swing as a pendulum, as shown. An electromagnet is mounted so that the ring passes over it as it swings.



The ring is set into oscillation with switch K open.

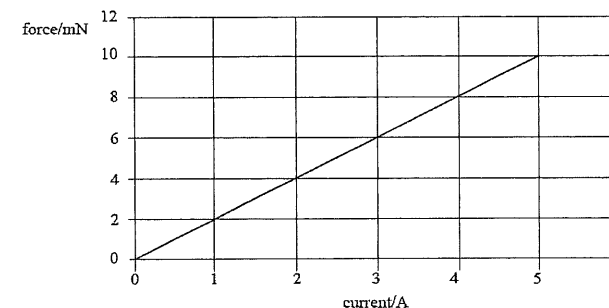
What happens to the motion after switch K has been closed?

- A The periodic time will decrease.
- B The oscillations will be damped.
- C The amplitude will increase because the ring is accelerated towards the magnet.
- D The oscillations will continue at constant amplitude while the battery can supply energy.

11. [NJC/H2/Prelims 2010/P1/32]

(Teacher's note: you need to assume that the part of the wire cutting the magnetic field vibrates with the same amplitude of 1.5 mm)

A wire of length 0.50 m, forming part of a complete circuit, is positioned at right angles to a uniform magnetic field. The graph shows how the force acting on the wire due to the magnetic field varies as the current in the wire is increased.



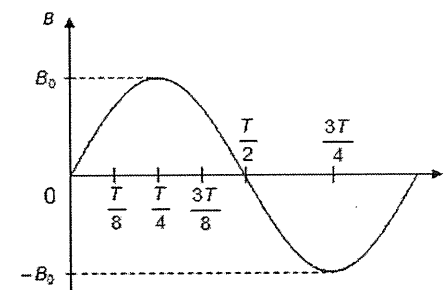
The circuit is switched off. The same piece of wire is then strummed such that it vibrates with an amplitude of 1.5 mm and at 100 Hz.

What is the maximum e.m.f. induced across the wire?

A 3.00×10^{-4} V B 1.88×10^{-3} V C 1.18 V D 1.88 V

12. [PJC/H2/Prelims 2010/P1/31]

A magnetic field is applied perpendicularly to the plane of a flat coil of copper wire. The time variation of the magnetic flux density is given by $B_0 \sin\left(\frac{2\pi}{T}t\right)$, as shown.

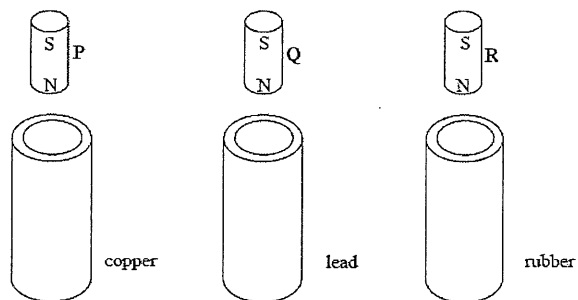


At which value of t is the magnitude of the e.m.f. induced in the coil a maximum?

A $\frac{T}{8}$ B $\frac{3T}{8}$ C $\frac{T}{2}$ D $\frac{3T}{4}$

13. [NJC/H2/Prelims 2010/P1/33]

Three vertical tubes, made from copper, lead and rubber respectively have identical dimensions. Identical, strong cylindrical magnets P, Q and R are released simultaneously from the same distance above each tube. The magnets are observed to emerge from the bottom of the tubes at different times.



Which row shows the correct order in which they will emerge?

Resistivity of copper = $1.7 \times 10^{-8} \Omega \text{ m}$

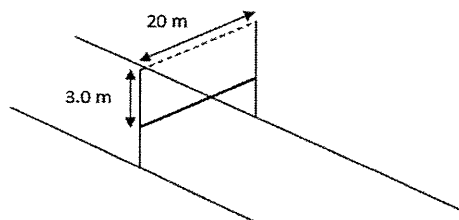
Resistivity of lead = $22 \times 10^{-8} \Omega \text{ m}$

Resistivity of rubber = $50 \times 10^{13} \Omega \text{ m}$

	emerges first	emerges second	emerges last
A	P	Q	R
B	R	P	Q
C	R	Q	P
D	P	R	Q

14. [PJC/H2/Prelims 2010/P1/32]

At the beginning of a horse-race, a horizontal straight wire of length 20 m is raised vertically through a height of 3.0 m in 0.20 s.

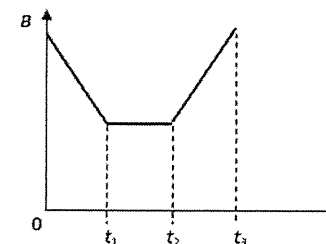


The horizontal component of the Earth's magnetic field strength perpendicular to the wire is $2.0 \times 10^{-5} \text{ T}$. What is the average e.m.f. induced across the ends of the wire?

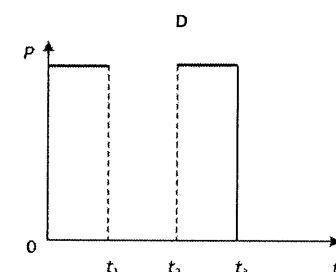
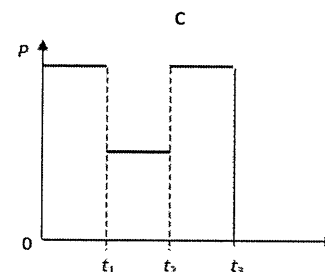
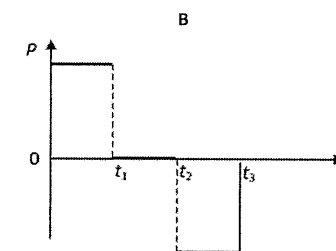
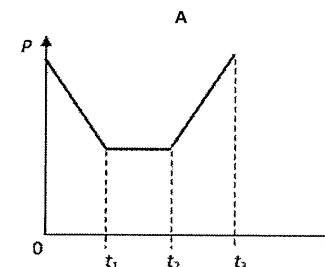
- A Zero B 0.24 mV C 1.2 mV D 6.0 mV

15. [NYJC/H2/Prelims 2010/P1/32]

A coil of wire of one turn only with a negligible resistance is connected to a resistor. A magnetic field B whose magnitude changes with time t as shown in the graph is applied perpendicularly to the coil.

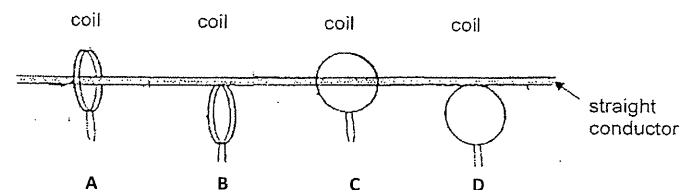


Which graph best shows the variation of the heating power P in the resistor with time t ?



16. [RI/H2/Prelims 2010/P1/31]

Large alternating currents in a straight conductor can be measured by the e.m.f. induced in a small coil. Which arrangement of the coil induces the largest e.m.f.?



17. [RI/H2/Prelims 2010/P1/32]

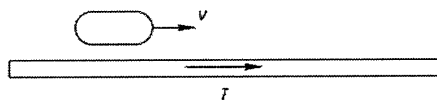
A right-hand drive car heads East at a speed of 20 m s^{-1} . It cuts the vertical component of the Earth's magnetic field of flux density $5.0 \times 10^{-5} \text{ T}$ acting downwards. The width of the car's bonnet is 1.5 m .

What is the e.m.f. generated across the bonnet and which side of the car will be positive?

	e.m.f. generated	side of car which is positive
A	0.67 mV	driver
B	0.67 mV	passenger
C	1.5 mV	driver
D	1.5 mV	passenger

18. [RVHS/H2/Prelims 2010/P1/30]

A loop of wire moves at speed v parallel to a long straight conductor carrying current I as shown.

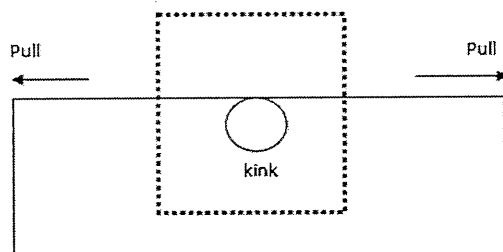


Which statement is correct?

- A The direction of the induced current in the loop is clockwise.
- B The magnitude of the induced current in the loop varies with the speed at which the loop moves.
- C The magnitude of the induced current in the loop varies with the magnitude of the current in the long straight conductor.
- D There is no induced current.

19. [SAJC/H2/Prelims 2010/P1/22]

A closed insulated wire with a circular kink is placed in a uniform magnetic field with a flux density of $2.5 \times 10^{-2} \text{ T}$, directed into the page. The diameter of the kink is 1.5 cm . The wire is quickly pulled taut and the kink is straightened in a time of 0.050 s . The wire resistance is 1.6Ω .

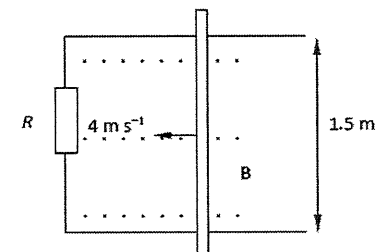


What is the power generated in the wire?

- A $3.3 \times 10^{-10} \text{ W}$
- B $4.9 \times 10^{-9} \text{ W}$
- C $7.7 \times 10^{-8} \text{ W}$
- D $8.8 \times 10^{-7} \text{ W}$

20. [RVHS/H2/Prelims 2010/P1/31]

A metallic rod slides along a frictionless wire frame of width 1.5 m with a speed of 4.0 m s^{-1} across a magnetic field of field strength 5.0 T perpendicular to the plane of the frame as shown.



What is the direction of the induced current and the power dissipated by the resistor R of resistance 12Ω ?

	direction of induced current	power dissipated
A	clockwise	2.5 W
B	anti-clockwise	2.5 W
C	clockwise	75 W
D	anti-clockwise	75 W

21. [VJC/H2/Prelims 2010/P1/28]

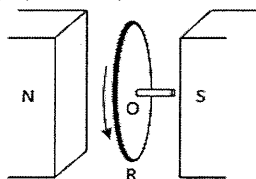
A flat circular coil of 120 turns, each of area 0.070 m^2 , is placed with its plane perpendicular to a uniform magnetic field. The flux density of the field is changed steadily from 80 mT to 20 mT without a change in direction, over a period of 2.0 s .

What is the e.m.f. induced in the coil during this time?

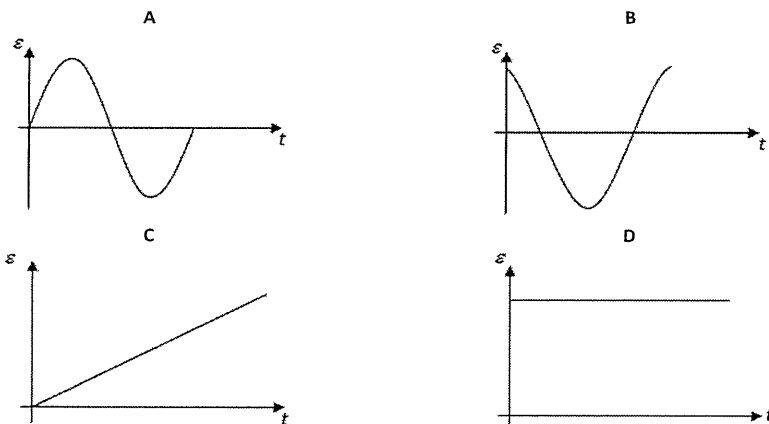
- A 0.13 V
- B 0.25 V
- C 0.42 V
- D 0.50 V

22. [TJC/H2/Prelims 2010/P1/31]

The figure below shows a copper disc rotating steadily about its centre O in a uniform magnetic field between two bar magnets. The magnetic field is acting perpendicularly to the disc.

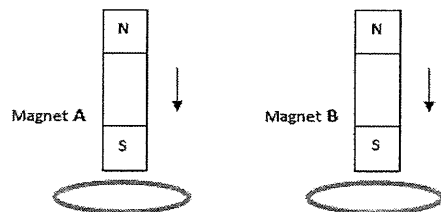


Which graph correctly shows the variation of the induced e.m.f. ε between the centre O and a point R on the rim of the disc with time t ?



23. [YJC/H2/Prelims 2010/P1/28]

Two identical magnets A and B are dropped from rest through the middle of a conducting ring. The ring for B has a small break in it as shown.

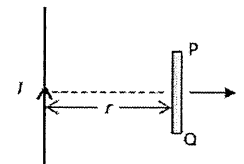


Which statement is correct?

- A Magnet A has a smaller acceleration.
- B Magnet B has a smaller acceleration.
- C Both magnets fall through with the same acceleration.
- D Both magnets fall through at constant velocity.

24. [TJC/H2/Prelims 2010/P1/32]

The diagram below shows a wire carrying a current I . The magnetic flux density due to the straight wire is given by $B = \frac{\mu_0 I}{2\pi r}$. A straight conductor PQ is placed in the same vertical plane as the wire and is moved at constant speed v away from the wire.

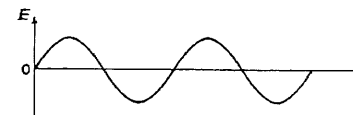


How will the magnitude of the induced e.m.f. in PQ vary and which end will be at a higher potential?

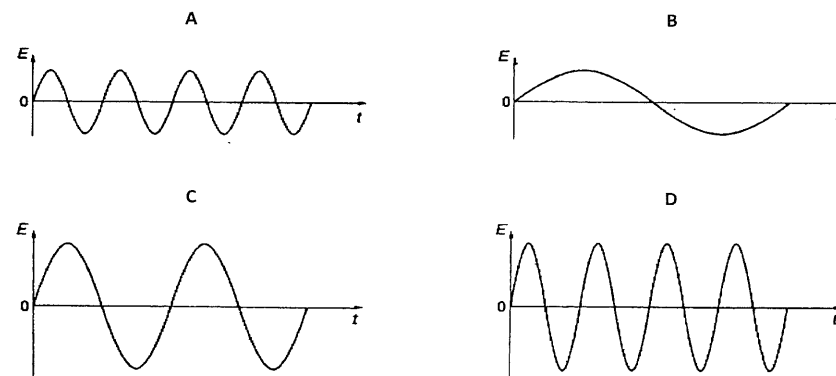
	magnitude of induced e.m.f.	end at higher potential
A	decreases	P
B	decreases	Q
C	increases	P
D	increases	Q

25. [YJC/H2/Prelims 2010/P1/29]

When a coil is rotated in a magnetic field, the induced e.m.f. E varies as shown.

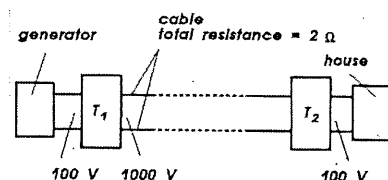


Which graph, drawn to the same scale, would be obtained if the speed of rotation of the coil were doubled?



26. [Dunman High/H2/Prelims 2010/P1/34]

In the figure below, a windmill drives a 100 V generator which produces electrical energy at a rate of 1000 W. This energy is supplied to a distant house through two cables of total resistance $2.0\ \Omega$. T_1 and T_2 are two transformers of 100% efficiency used to step up and step down the voltage respectively.

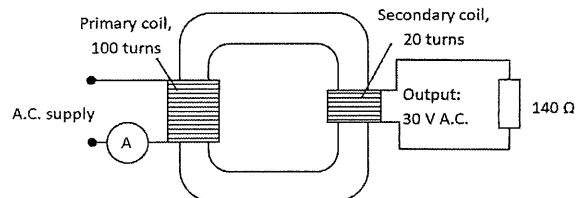


What is the rate of energy dissipated in the cables?

- A 1.0 W B 2.0 W C 20 W D 100 W

27. [HCI/H2/Prelims 2010/P1/32]

A 100% efficient transformer is connected as shown to a sinusoidal a.c. supply.



What is the reading on the ammeter?

- A 0.0429 A B 0.214 A C 1.07 A D 2.14 A

28. [IJC/H2/Prelims 2010/P1/33]

A transformer has N_p turns in its primary coil and N_s turns in the secondary coil, and there is no flux leakage. The alternating voltage and current in the primary coil are V_p and I_p respectively. The alternating voltage and current in the secondary coil are V_s and I_s respectively.

Which one of the following relations **must** be correct?

- A $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_s}{I_p}$ B $\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$ C $\frac{N_s}{N_p} = \frac{V_s}{V_p}$ D $\frac{N_s}{N_p} = \frac{I_p}{I_s}$

29. [NYJC/H2/Prelims 2010/P1/31]

The iron cores of transformers are usually 'laminated', that is, the iron core is constructed of thin strips.

What is the principal purpose of this lamination?

- A To reduce eddy currents in the core.
B To increase the magnetic field strength in the core.
C To reduce the electrical resistance of the core.
D To improve the flux linkage of the two coils through the core.

30. [RI/H2/Prelims 2010/P1/34]

A transformer steps up 120 V at the primary coil to 240 V at the secondary coil.

If the current in the primary coil is 2.0 A and the power loss in the windings and core of the transformer is 48 W, what is the current in the secondary coil?

- A 0.20 A B 0.80 A C 1.0 A D 1.2 A

31. [SAJC/H2/Prelims 2010/P1/31]

In an ideal transformer, the most important function of the soft-iron core is

- A to reduce eddy-currents.
B to improve the flux-linkage between the primary and secondary coils.
C to dissipate the heat generated by the two coils.
D to produce a uniform magnetic field in the two coils.

32. [TJC/H2/Prelims 2010/P1/33]

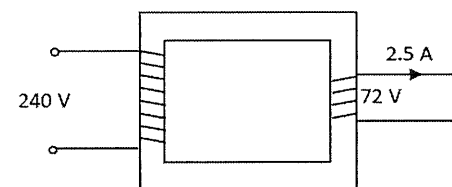
A generator produces a current of 60 A r.m.s. at a voltage of 120 V r.m.s. The voltage is stepped up to 4500 V r.m.s. by an ideal transformer and transmitted through a power line of total resistance $1.0\ \Omega$.

What percentage of the power generated is lost in the transmission line?

- A 0.018 % B 0.024 % C 0.036 % D 0.048 %

33. [VJC/H2/Prelims 2010/P1/31]

A 240 V mains voltage is applied to the primary coil of a step-down transformer. A resistor is placed across the secondary coil, drawing a current of 2.5 A from the secondary coil, which has a voltage of 72 V across it.



What is the current flowing in the primary coil?

- A 0.53 A B 0.75 A C 1.1 A D 8.3 A

II. Structured Questions

1. [AJC/H2/Prelims 2010/P3/3] (Teacher's note: You need to assume the wire frame is massless)

a. A simple current balance is shown in Fig. 1.1.

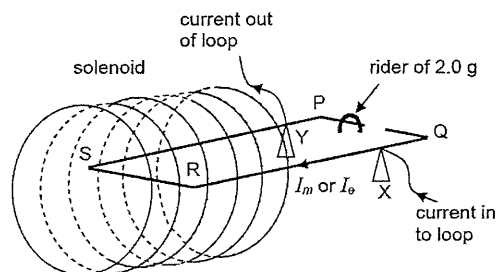


Fig. 1.1

A solenoid is connected to a horizontal rectangular copper loop PQRS, such that the same current can pass through them. The loop is pivoted on two knife-edges X and Y and a rider of mass 2.0 g is placed on PQ. Electrical connections are made through the knife-edges.

Fig. 1.2 shows how the magnetic flux density B varies along the length of the solenoid.

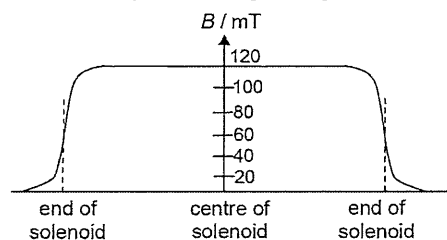


Fig. 1.2

When SR is at the centre of the solenoid and a current I_m is passed through the loop, XY needs to be at a quarter length of PS from PQ to balance the loop. When SR is at the end of the solenoid and a current I_e is passed through the loop, XY needs to be shifted to midway of the loop in order to regain its balance.

- Define *magnetic flux density*. [1]
- On Fig. 1.1, indicate the direction of the current through the solenoid. Label it I_s . Explain how you have obtained your answer. [2]
- Determine the ratio of I_m to I_e . [3]

- b. Fig. 1.3 shows a rectangular coil placed at the centre of the solenoid with its plane perpendicular to the axis of the solenoid.

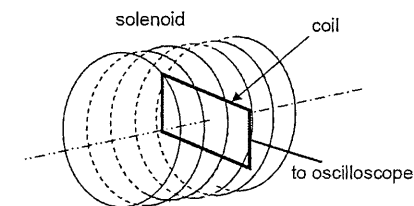


Fig. 1.3

The solenoid has a cross-sectional area of 0.0050 m^2 and length 50 cm, having 400 turns. The rectangular coil has 8 turns with dimensions of 0.010 m by 0.020 m . The ends of the coil are connected to a cathode ray oscilloscope.

An alternating current of 50 Hz is passed through the solenoid. Its variation with time is as shown in Fig. 1.4.

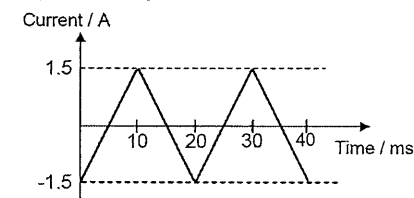


Fig. 1.4

- Explain why an e.m.f. will be induced in the coil. [2]
- Calculate the e.m.f. induced in the coil. [3]
- On Fig. 1.5, sketch how the induced e.m.f. in the coil varies with time over two cycle of current change. [4]

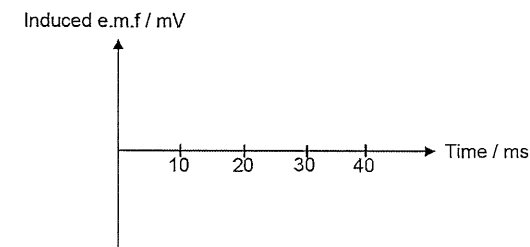


Fig. 1.5

2. [Dunman High/H2/Prelims 2010/P3/7]

Fig. 2.1 shows a ring falling from rest through a coil A carrying a constant current.

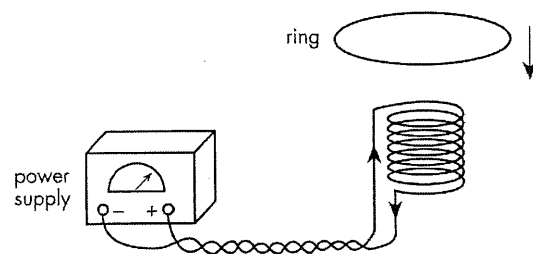
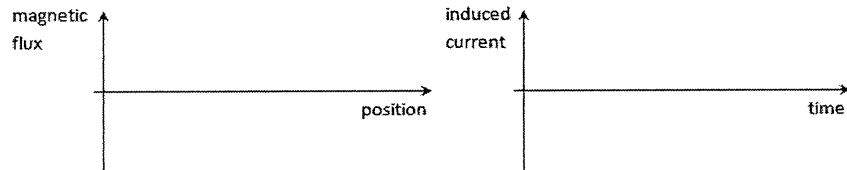


Fig. 2.1

- a. i. List two factors which will affect the magnitude of e.m.f. induced in the ring. [2]
- ii. Using the appropriate laws of electromagnetism, state and explain how the magnitude and direction (as viewed from the top) of the current in the ring changes throughout the motion in which the ring approaches, falls through and away from coil A. [6]
- iii. On the separate axis below, sketch graphs to show the variation of [2]
1. the magnetic flux through the ring with position of the ring
 2. the induced current in the ring with time



- iv. If air resistance is ignored, state and explain whether the ring falls through the coil A with a constant acceleration. [2]
- b. The constant current source connected to coil is now replaced by a varying e.m.f. which produces a current in the coil A as shown in Fig. 2.2.

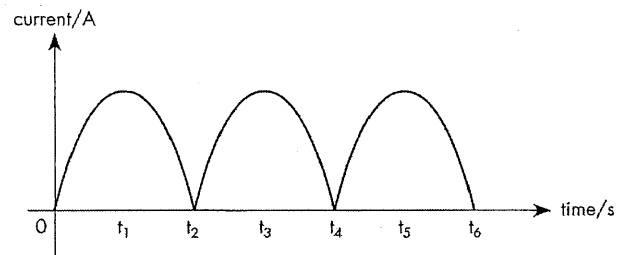


Fig. 2.2

In addition, coil A is wound on one side of an iron core with coil B on the other side as shown in Fig. 2.3.

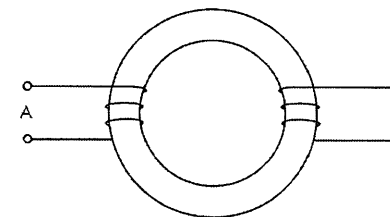
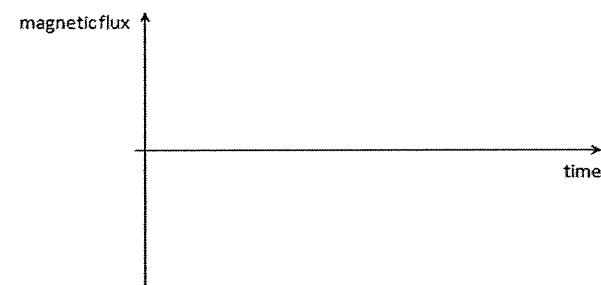
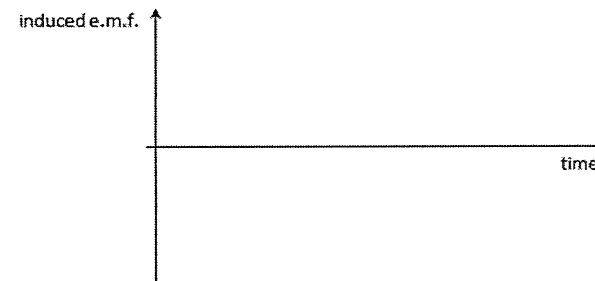


Fig. 2.3

- i. Sketch a graph to show the variation of magnetic flux produced by the current in coil A with respect to time. [2]



- ii. Sketch a graph to show the variation of e.m.f. induced across coil B with respect to time. [2]



[Total: 16]

3. [MJC/H2/Prelims 2010/P3/4]

- a. State *Faraday's law of electromagnetic induction*. [1]
- b. A long straight wire carries a direct current. A rigid loop of conducting wire is placed near the wire such that the wire is in the plane of the loop. The loop is moved at constant speed away from the wire as shown in Fig. 3.1.

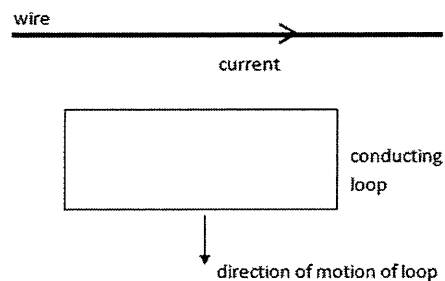


Fig. 3.1

- i. Explain why an e.m.f. is induced in the loop. [2]
- ii. On Fig. 3.1, draw an arrow to indicate the direction of the current induced in the loop and explain your answer. [2]
- iii. It was found that energy is dissipated in the wire loop. Explain how the movement of the loop gives rise to energy dissipation. [3]

[Total: 8]

4. [NJC/H2/Prelims 2010/P2/4]

- a. State the laws of electromagnetic induction. [2]
- b. An accelerometer in a vehicle consists of a steel bar mounted on two flexible blades. The bar passes through two coils, as shown in Fig. 4.1. An alternating voltage is applied to the primary coil.

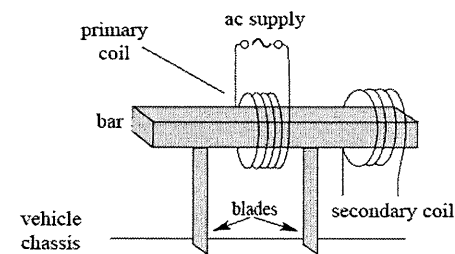


Fig. 4.1

- i. Explain why a voltage is induced in the secondary coil. [2]
- ii. Explain why the bar is displaced from its equilibrium position when the vehicle accelerates in a direction parallel to the axis of the bar, as shown in Fig. 4.2. [2]

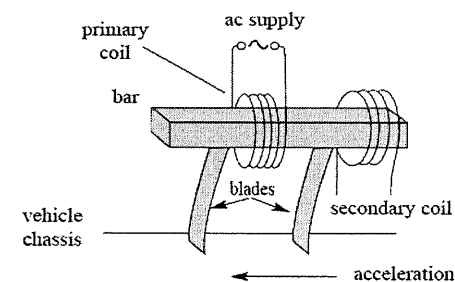


Fig. 4.2

- iii. Explain why the amplitude of the secondary voltage changes when the bar is displaced from its equilibrium position in a direction along the axis of the bar. [2]

[Total: 8]

5. [NYJC/H2/Prelims 2010/P3/4]

Fig. 5.1 shows the cross section of the mechanism of a coin-operated vending machine.

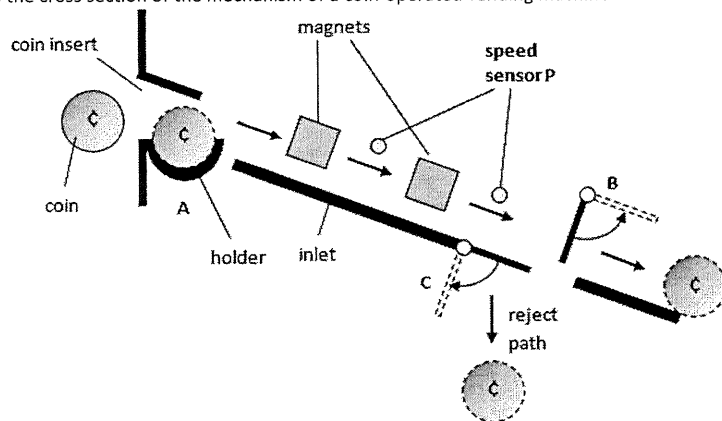


Fig. 5.1

On entering the vending machine at A, the holder will stop the coin momentarily before releasing it into the inlet.

At the inlet, the coin will pass through 2 pairs of magnets and 2 pairs of speed sensors where eddy current will be generated in the coin. The coin travels down the inlet more slowly due to the presence of the magnet as compared to an inlet without the pair of magnets. If the speed sensors indicate that the coin has the correct speed, the gate at B will open and accept the coin. Otherwise, the gate at C will open and the coin will be rejected.

- State *Faraday's law of electromagnetic induction*. [1]
- Use Faraday's Law of electromagnetic induction to explain the generation of eddy current as the coin travels down the inlet. [3]
- Using the law of conservation of energy, explain why the magnets cause the coin to travel down more slowly as compared to an inlet without the pair of magnets. [2]
- Describe how this mechanism can help to detect the correct denomination of coins. [2]

[Total: 8]

6. [RI/H2/Prelims 2010/P2/6]

Fig. 6.1 shows the front view of a large flat circular coil connected to a sinusoidal alternating voltage supply and a small flat circular coil placed at the centre of the large coil such that the planes of the two coils are coincident. The smaller coil is connected to a cathode-ray oscilloscope (c.r.o.).

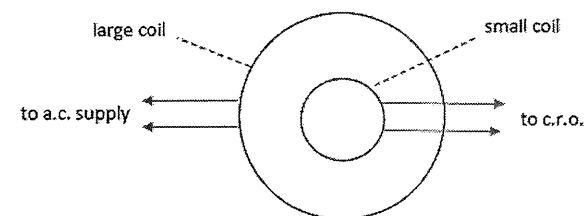


Fig. 6.1

- If the variation of the sinusoidal alternating current to the large coil is as shown in Fig. 6.2, draw sketch graphs, one in each case, to show the variation with time of [2]
 - the magnetic flux through the small coil
 - the induced e.m.f. in the small coil as displayed on the c.r.o..

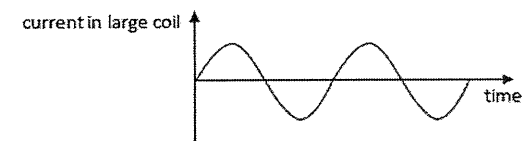
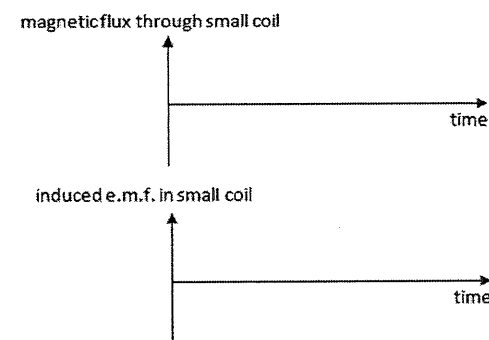


Fig. 6.2



- Justify the shape of your sketches in a.. [3]
- State and explain how the trace on the screen of the c.r.o. would be affected if the small coil is rotated such that the angle between the planes of the two coils increase from zero to 90° whilst maintaining a constant root mean square current in the large coil. [2]

[Total: 7]

7. [RVHS/H2/Prelims 2010/P3/6]

Fig. 7.1 shows a system used by an engineer to determine the rate of revolution of a rotating axle.

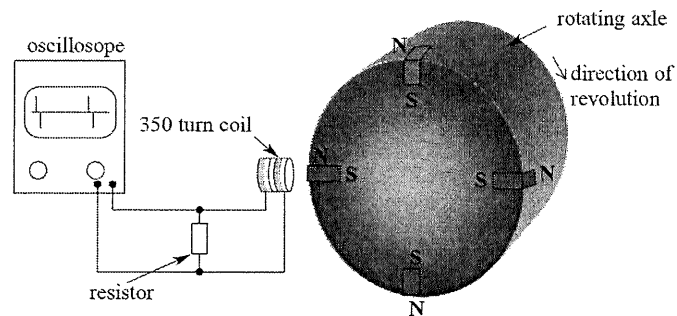


Fig. 7.1

Four small bar magnets are embedded in the axle as shown. The N pole of each magnet is towards the outside of the axle. A voltage is produced between the terminals of a coil placed close to the rotating axle. The voltage produced is monitored using an oscilloscope. The waveform produced is shown in Fig. 7.2.

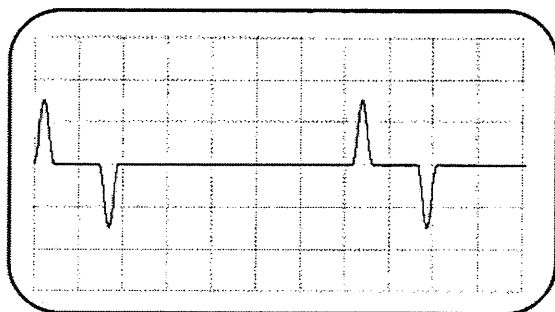


Fig. 7.2

The Y gain setting = 5 mV div^{-1}

The time-base setting = 10 ms div^{-1}

- Determine the number of revolutions made by the axle in one minute. [3]
- State the laws of electromagnetic induction and use them to explain the shape of the voltage pulses produced. [6]
 - The coil has 350 turns. Determine the maximum rate of change of flux through the coil and state its units. [4]
- State and explain what would be observed if the direction of revolution is changed. [2]
- On Fig. 7.2, draw the waveform that shows the changes you would expect to see when the rate of revolution of the axle is doubled. Label the waveform E . [3]
- It is often said that the very act of measuring affects the measurement itself. Comment on this statement with reference to your answer in a.. [2]

[Total: 20]

8. [SAJC/H2/Prelims 2010/P2/4]

- Define *magnetic flux density*. [1]
- A magnetohydrodynamic drive (MHD) is a method for propelling seagoing vessels using only electric and magnetic fields. Sea water is propelled through the MHD tube using magnetic forces. Fig. 8.1 shows the components of the MHD. A pair of powerful magnet is placed across the MHD tube where a current flows.

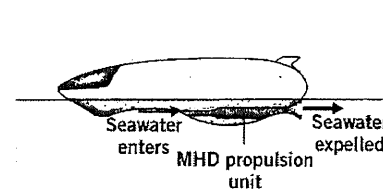


Fig. 8.1

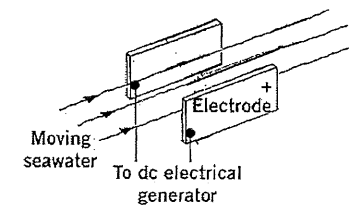


Fig. 8.2

- On Fig. 8.2, draw an arrow to indicate the direction of the magnetic field that allows the MHD to propel water in the direction shown. [1]
 - Calculate the magnetic flux density required to produce a force of 16000 N with a current of 10.0 A . The separation between the electrode is 30.0 cm . [2]
 - Suggest one advantage and one disadvantage of MHD. [2]
- c. A bar magnet attached to a spring falls vertically with its south pole entering a horizontal closed metallic wire loop as shown in Fig. 8.3.

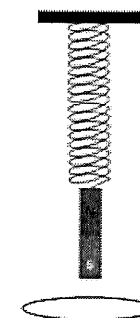


Fig. 8.3

- Sketch a graph of the variation of the induced current in the loop with time as the magnet passes through the loop. Draw at least 2 cycles of the change. [2]
- Explain the features of the graph in part i.. [2]

[Total: 10]

9. [TJC/H2/Prelims 2010/P3/3]

A current-carrying solenoid XY is placed near to a small coil, as shown in Fig. 9.1.

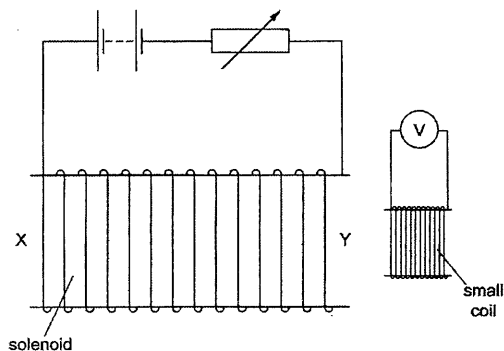


Fig. 9.1

The small coil is connected to a sensitive voltmeter.

- Explain what is meant by *magnetic flux linkage of the coil*. [1]
- On Fig. 9.1, draw an arrow to indicate the direction of the magnetic field inside the solenoid. [1]
- The resistance of the variable resistor is changed so that the current in the solenoid increases. Using the laws of electromagnetic induction, explain why a reading is recorded on the voltmeter. [3]
- The current in the solenoid is now made to vary as shown in Fig. 9.2.

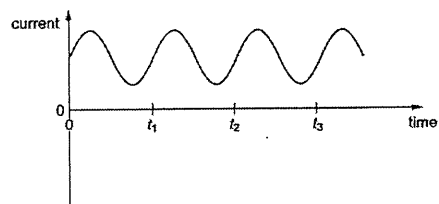


Fig. 9.2

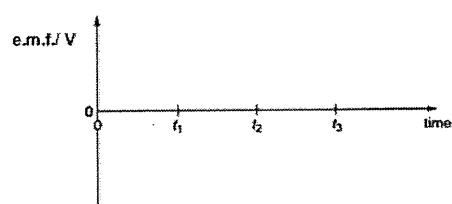


Fig. 9.3

Sketch the variation with time of the e.m.f. induced in the small coil in Fig. 9.3.

[2]

[Total: 6]

10. [VJC/H2/Prelims 2010/P2/2(modified)]

- State *Faraday's law of electromagnetic induction*. [1]
- A plane coil of wire containing 50 turns each of diameter 5.0 cm and total resistance 5.0Ω is placed so that the plane of the coil is normal to a uniform magnetic field B of flux density $2.0 \times 10^{-3} \text{ T}$. The coil is moved with a speed of 15.0 m s^{-1} , at right angles to the magnetic field from P to the positions Q and R shown dotted in Fig. 10.1.

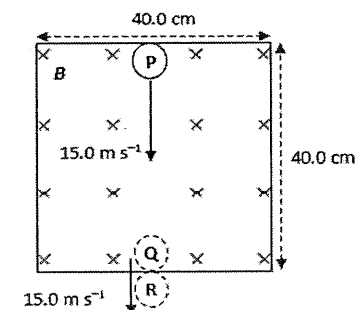


Fig. 10.1

Calculate the average induced e.m.f. in the coil as it travels from

- P to Q [1]
 - Q to R. [3]
- c. The coil is now replaced by a semi-circular wire MN with the same diameter of 5.0 cm and moved with a speed of 15.0 m s^{-1} at right angles to the same magnetic field as shown in Fig. 10.2.

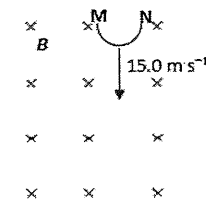


Fig. 10.2

- Determine the magnitude of the induced e.m.f. generated between M and N. [2]
- State whether M or N is at a higher potential. [1]

[Total: 8]

11. [YJC/H2/Prelims 2010/P2/4]

A circular coil of diameter 140 mm has 850 turns. It is oriented so that its plane is perpendicular to a horizontal magnetic field of uniform flux density 45 mT, as shown in Fig. 11.1.

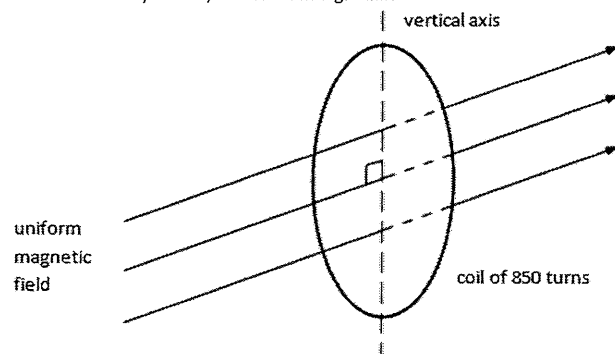


Fig. 11.1

- a. Calculate the magnetic flux passing through the coil in this position. [2]
- b. The coil is rotated through 90° about the vertical axis shown in a time of 120 ms.
 - i. Calculate
 1. the change of magnetic flux linkage produced by this rotation [2]
 2. the average e.m.f. induced in the coil during this rotation. [1]
 - ii. State and explain what will happen to the value of the average e.m.f. induced if the coil is rotated through 360° . [2]

[Total: 7]

12. [MI/H2/Prelims 2010/P3/6 (part)]

Fig. 2.1 shows 2 coils X and Y wound on a soft iron core.

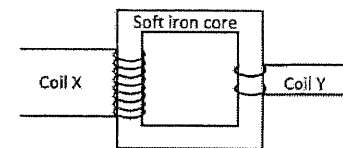


Fig. 2.1

- a. Magnetic flux links coils X and Y when there is a current in coil X.
 - i. What happens in coil Y when the magnetic flux in the soft iron core changes? [1]
 - ii. State a law that justifies your answer in a.i.. [1]
- b. The output of coil Y is connected to a 12 V, 3.0 A light bulb. The input of coil X is now connected to a 240 V r.m.s., 100 Hz mains supply. (Assume that the efficiency of energy transfer between the two coils is 100% and the bulb lights with full intensity.) Calculate the current in coil X. [2]
- c. Sketch a graph to show how the power dissipated in the light bulb connected to coil Y varies with time over one cycle of the alternating output voltage. [2]

[Total: 6]

13. [MJC/H2/Prelims 2010/P2/3]

From the power station, the voltage is stepped up to about 230 kV along the high voltage cables before a series of step down transformers near homes are used to reduce the voltage to 240 V at our homes. Fig. 3.1 shows the last iron-cored step down transformer before our homes with the actual turns ratio indicated. The home 240 V output has a frequency of 50 Hz is connected to a $58\ \Omega$ resistor of a home appliance.

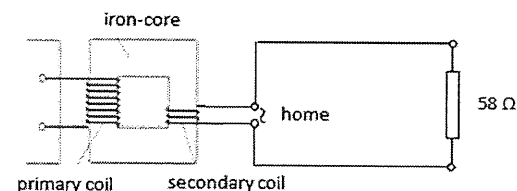


Fig. 3.1

- a. Determine the number of similar turns-ratio transformers required to step down from the high voltage (230 kV) cables to the 240 V output used at homes. State one assumption made. [4]
- b. Explain the purpose of the iron core in the transformer. [1]
- c. Determine the average power dissipated in the $58\ \Omega$ resistor of a home appliance. [2]
- d. Suggest a typical household appliance with such power rating. [1]

[Total: 8]

- 13 C As the magnets enter the tubes, emf and hence current is induced in the tubes. According to Lenz's law, the direction of the induced current would oppose the motion of the magnets. A tube with a larger value of resistivity would have a lower induced current and hence there is less opposition to the motion of the magnet falling through this tube and it would emerge faster than one falling through a tube with a lower value of resistivity.

Hence, the magnet falling through the rubber tube would emerge first, followed by that falling through the lead tube and finally the copper tube.

14 D $|E| = \left| \frac{d\Phi}{dt} \right| = B \frac{dA}{dt} = (2.0 \times 10^{-5}) \frac{(3.0 \times 20)}{0.20} = 6.0 \times 10^{-3} \text{ V} = 6.0 \text{ mV}$

15 D $E = -\frac{d\Phi}{dt} \propto -\frac{dB}{dt}$

From 0 – t_1 , constant positive E induced

From t_1 to t_2 , zero E induced

From t_2 to t_3 , constant negative E induced

Magnitude of induced E from 0 to t_1 = Magnitude of induced E from t_2 to t_3

$$P = \frac{E^2}{R}$$

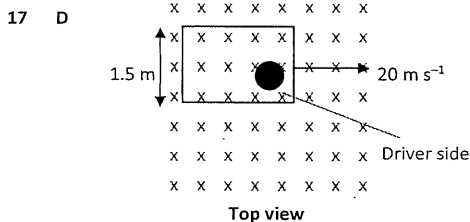
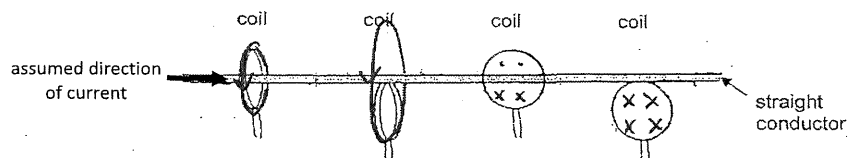
Constant positive P of the same magnitude for 0 to t_1 and t_2 to t_3

Zero P for t_1 to t_2 .

- 16 D **Coil A & Coil B:** No flux passing perpendicularly through the coil. Magnetic flux around the conductor is in the same plane as the coil.

Coil C: equal amount of flux passing out of the plane of the coil and into the plane of the coil. No net flux linkage.

Coil D: Maximum flux linkage through the plane of the coil, hence largest e.m.f. induced.



$$E = BLv = (5.0 \times 10^{-5})(1.5)(20) = 1.5 \text{ mV}$$

Using Fleming's Right Hand rule, current induced would flow from the driver to the passenger side. Since the bonnet acts as an e.m.f. source, the passenger side would have higher potential.

- 18 D There is no change in flux linkage through the loop as it moves along the conductor. Hence, no e.m.f. and current is induced.

- 19 B Emf is induced when the flux linkage through the circular kink decreases.

$$|E| = \left| \frac{d\Phi}{dt} \right| = B \frac{dA}{dt} = (2.5 \times 10^{-2}) \frac{\left(\pi \times \left[\frac{0.015}{2} \right]^2 \right)}{0.050}$$

$$P = \frac{E^2}{R} = \left[(2.5 \times 10^{-2}) \frac{\left(\pi \times \left[\frac{0.015}{2} \right]^2 \right)}{0.050} \right]^2 / 1.6 = 4.9 \times 10^{-9} \text{ W}$$

20 D $P = \frac{E^2}{R} = \frac{(BLv)^2}{R} = \frac{(5.0 \times 1.5 \times 4.0)^2}{12} = 75 \text{ W}$

Using Fleming's right hand rule, the induced current flows from the lower end to the upper end of the metal rod. Hence, current will flow in an anticlockwise direction through the circuit.

21 B $|E| = \left| \frac{d\Phi}{dt} \right| = \left| NA \frac{dB}{dt} \right| = (120)(0.070) \left(\frac{60 \times 10^{-3}}{2.0} \right) = 0.25 \text{ V}$

- 22 D E.m.f. induced is directly proportional to the rate of rotation of the disc. Since the disc is rotating at a constant rate (constant ω), e.m.f. induced is constant.

- 23 A As magnet A falls through a closed ring, a current flowing in a direction opposing the motion of the magnet will be induced in the ring, causing it to slow down.
As magnet B falls through a ring with a small break in it, there will be no current induced in the ring as it is not a closed circuit.

- 24 A The magnetic flux density B of the wire decreases with distance r from the wire. Since $E = BLv$, e.m.f. induced in PQ will decrease as PQ is moved away from the wire at a constant speed.
Using Fleming's right hand rule, the induced current would flow from Q to P. Since PQ acts as a source of e.m.f., the end P will have a higher potential.

25 D $E = NBA\omega \sin \omega t$

Doubling the rate of rotation ω would double both the maximum induced e.m.f. and double the frequency of the induced e.m.f.

- 26 B 100% efficiency transformer T1 implies, electrical power generated = electrical power transmitted to cables
 $1000 = I_c V_c = I_c \times 1000$
 $\Rightarrow$ current through cables $I_c = 1 \text{ A}$
power dissipated in cables = $12 \times 2.0 = 2.0 \text{ W}$

27 A $V_p = \frac{N_p}{N_s} V_s = 150 \text{ V}$

100% efficiency transformer implies, electrical power generated at primary coil = electrical power transmitted

to secondary coil = $\frac{30^2}{140} = 6.43 \text{ W}$

$I_p \times 150 = 6.43$

$I_p = 0.0429 \text{ A}$

- 28 C Since the transformer may not be ideal, its efficiency in electrical power transfer from primary coil to secondary coil is less than 100%.

This implies that $I_p V_p > I_s V_s$. Hence only C is correct since there is no flux leakage.

- 29 A Lamination is to reduce the magnitude of eddy currents by restricting the eddy currents to circulate in very small volumes of the iron core. This greatly reduces the lost in electrical energy as heat through Joule heating.

- 30 B Power generated at primary coil – power loss in windings = Power transmitted to secondary coil

$I_p V_p - P_{\text{loss}} = I_s V_s$

$120 \times 2 - 48 = I_s \times 240$

$I_s = 0.80 \text{ A}$

- 31 B Soft iron is used in a transformer as it has a high magnetic permeability and is able to concentrate magnetic fields/flux that links the primary coil to the secondary coil, minimizing the loss in the flux linkage.

- 32 C Mean power generated, $\langle P_{\text{in}} \rangle = 60 \times 120 = 7200 \text{ W}$

Since transformer is ideal, $\langle P_{\text{in}} \rangle = \langle P_{\text{out}} \rangle \Rightarrow 4500 \times I_{\text{cables}} = 7200 \Rightarrow I_{\text{cables}} = 1.6 \text{ A}$

% power lost in cables = $\frac{1.6^2 \times 1.0}{7200} \times 100 = 0.036\%$

- 33 B For an ideal transformer $I_p V_p = I_s V_s \Rightarrow \frac{I_p}{I_s} = \frac{V_p}{V_s}$

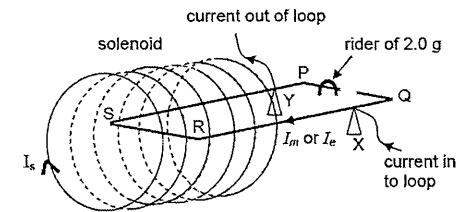
$I_p = \frac{2.5}{240} \times 72 = 0.75 \text{ A}$

Structured Questions

1. [AJC/H2/Prelims 2010/P3/3]

- a. i. Refer to definitions in your lecture notes.

ii.



For equilibrium, magnetic force acts perpendicularly downwards on SR.

By Fleming's Left Hand Rule, magnetic flux density is directed along axis of solenoid in direction of S to P. From Right Hand Grip Rule, current flows as indicated.

- iii. Let h be the length of PS, L be the length of SR, and W be the weight of the rider.

Consider equilibrium of the loop when SR is at the centre of the solenoid:

$$B_m I_m L \left(\frac{3}{4} h \right) = W \left(\frac{1}{4} h \right) \quad (1)$$

Consider equilibrium of the loop when SR is at the end of the solenoid:

$$B_e I_e L \left(\frac{1}{2} h \right) = W \left(\frac{1}{2} h \right) \quad (2)$$

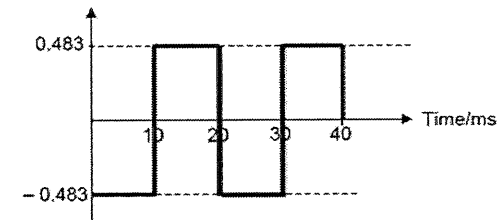
$$(1)/(2): \frac{B_m I_m}{B_e I_e} \times \frac{3}{2} = \frac{1}{2} \Rightarrow \frac{I_m}{I_e} = \frac{1}{3} \frac{B_e}{B_m} = \frac{1}{3} \times \frac{60}{120} = 0.167$$

- b. i. Alternating current causes a changing magnitude of the magnetic flux density within the solenoid.

Changing magnetic flux density passing through the coil causes a rate of change of magnetic flux linkage through the coil. This induced an e.m.f.

$$\begin{aligned} E &= \frac{d\Phi}{dt} = \frac{d}{dt} (NBA) = NA \frac{dB}{dt} = NA \mu_0 n \frac{dI}{dt} \\ &= 8 \times (0.010) \times (0.020) \times \mu_0 \times \frac{400}{50 \times 10^{-2}} \times \left[\frac{1.5 \text{ A} - (-1.5 \text{ A})}{10 \times 10^{-3} \text{ s}} \right] = 4.83 \times 10^{-4} \text{ V} \end{aligned}$$

- iii. Induced e.m.f. / mV



2. [Dunman High/H2/Prelims 2010/P3/7]

a. i. Any two:

- area of the ring
- magnitude of magnetic field / current in the coil
- orientation of the coil
- orientation of the ring
- number of turns in coil A

ii. As the ring approaches the coil, there is increase in magnetic flux linkage through the ring and by Faraday's law, an e.m.f. is induced.

The magnitude of current increases from zero and it flows in an anti-clockwise direction around the ring so as to set up a magnetic field to oppose the increase in flux linkage according to Lenz's Law.

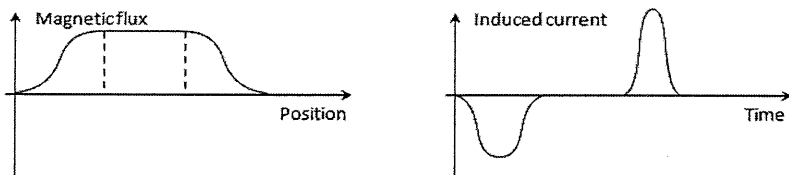
When the ring falls through the coil, there is no change in magnetic flux linkage through the ring and hence no induced current flows in the ring.

When the ring falls below the coil, there is decrease in magnetic flux linkage through the ring (and by Faraday's law, an e.m.f. is induced).

The induced current flows in a clockwise direction (to oppose the decrease in flux linkage according to Lenz's Law).

When the ring is at a great distance from the coil, there is no change in magnetic flux linkage through the ring and hence no induced current flows in the ring.

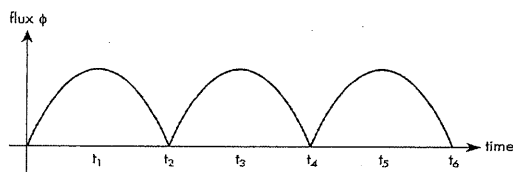
iii.



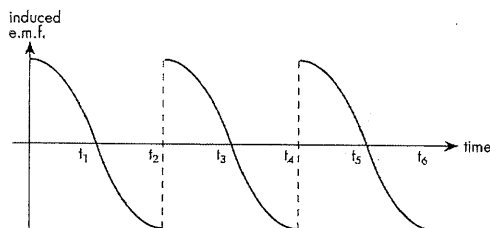
iv. Since the induced current changes with time, the magnitude of the magnetic force exerted on the ring changes with time.

Thus the ring falls with non-uniform acceleration.

b. i.



ii.



3. [MJC/H2/Prelims 2010/P3/4]

a. Faraday's Law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.

b. i. Changing flux linkage approach:

The current in the wire generates a magnetic field that decreases with distance from it ($B \propto 1/r$). Thus the magnetic flux linkage through the loop will be less further away from the loop.

According to Faraday's law, when the magnetic flux linkage through the loop changes with time, e.m.f. is induced in the loop.

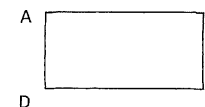
OR

Cutting of flux approach:

The current in the wire generates a magnetic field that decreases with distance from it ($B \propto 1/r$).

As the loop moves away, the sides AB and DC cut the magnet flux and thus generate e.m.f. ($A \rightarrow B$, $D \rightarrow C$).

Since AB is nearer to the wire, it cuts flux at a greater rate than DC and generates more e.m.f. ($E_{AB} > E_{DC}$), hence there is a net e.m.f. that is able to drive current around the loop.



ii. Changing flux linkage approach:

Using Right Hand Grip Rule, the B field is pointing into the paper through the loop.

When the loop is moving downwards, the magnetic flux linkage directed into the paper through the loop is *decreasing*. According to Lenz's law, the induced current must produce a flux into the paper in order to oppose the decrease. Hence, the induced current flows clockwise.

Clockwise direction

OR

Cutting of flux approach:

Both AB and DC produce e.m.f. pointing to the right (A to B , D to C) when moving downwards, as determined by Fleming's Right Hand Rule.

(Motion + B -field $\rightarrow$ induced e.m.f.)

As $E_{AB} > E_{DC}$ (as determined in i.), there is net e.m.f. that drives a current around the loop in a clockwise direction.

iii. Movement of the loop away from the wire caused in a change in flux linkage that resulted in an induced e.m.f. and current around the closed loop.

This results in power dissipated $P = I^2 R$ as heat (Joule heating) as the conducting loop has some resistance.

The power is supplied by the external force that keeps the loop moving at constant speed, otherwise it would slow down.

(Another accepted explanation revolves around the work done to overcome the attractive forces that are set up between the loop and the current in the wire to keep it moving at constant speed.)

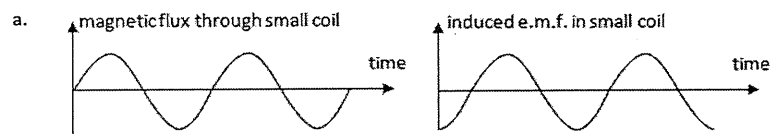
4. [NJC/H2/Prelims 2010/P2/4]

- a. Faraday's Law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage. Lenz's law states that the direction of the induced e.m.f. is such as to cause effects to oppose the change producing it.
- b. i. The alternating current in the primary coil produces an alternating magnetic flux in the bar which links to the secondary coil. Since there is a change in the magnetic flux linkage, there will be induced alternating voltage in the secondary coil.
- ii. The bar is displaced from its equilibrium position because inertia causes it to lag behind the other parts of the vehicle. When this happens, the blades are bent and this results in a net force acting on it that accelerate it together with the vehicle.
- iii. Since the magnetic flux in the bar is non-uniform, the magnetic flux in the secondary coil changes when bar is displaced. By Faraday's law, since the secondary voltage is proportional to rate of change of magnetic flux, the amplitude of secondary voltage changes.

5. [NYJC/H2/Prelims 2010/P3/4]

- a. Faraday's Law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
- b. As the coin moves down the inlet, it will approach and move away from the magnets, experiencing increase and decrease of magnetic flux. By Faraday's Law, an e.m.f. will be induced in the coin. As the coin is a conductor, the e.m.f. induced will induce an eddy current in the coin, which will produce an induced flux to oppose the change causing it.
- c. The generation of the eddy current generates electrical energy which will dissipate as heat in the coin. By conservation of energy, the electrical energy generated has been converted from the kinetic energy of the coin, thus, the coin will travel down more slowly as compared to an inlet without the pair of magnets.
- d. Different denomination of coins has different dimensions and may be made of different material. As such, the resistances of the coins are different which will affect the magnitude of the eddy current induced in the coins. This will affect the speed of the coins as they approach the speed sensors. By presetting quantities to correspond to various dimensions, we can differentiate the various denominations.

6. [RI/H2/Prelims 2010/P2/6]



- b. Magnetic flux through the small coil is proportional to the magnetic flux density which is proportional to the alternating current in the large coil. Hence the magnetic flux-time graph through the small coil has the same shape as that of the current-time graph for the large coil. Induced e.m.f. in the small coil is proportional to the rate of change of magnetic flux linkage, which is obtained from the negative of the slope of the magnetic flux- time graph.
- c. When the planes of the two coils are at 90° to each other, the magnetic field due to the current in the large coil is parallel to the plane of the small coil. Hence there is no magnetic flux linkage with the small coil and no e.m.f. will be induced in the small coil. The trace on the c.r.o. will show the amplitude of the induced e.m.f. reduced to zero.

7. [RVHS/H2/Prelims 2010/P3/6]

- a. Time interval between the 2 magnets passing through the coil: 71 to 76 ms
Thus, time for one revolution = $(71 \text{ to } 76 \text{ ms}) \times 4 = 284 \text{ to } 304 \text{ ms}$

$$\text{Hence, total number of revolutions per minute is: } \frac{60}{(284 \text{ to } 304) \times 10^{-3}} = 197 \text{ to } 212 \text{ rev min}^{-1}$$

- b. i. Faraday's Law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage. Movement of magnet changes flux (linkage) with coil
Voltage induced proportional to rate of change of flux (linkage)
Lenz's law states that the direction of the induced e.m.f. is such as to cause effects to oppose the change producing it.
Magnetic N-pole approaches coil, flux linkage increases and current is induced in a direction to repel the magnet.
Magnetic N-pole moves away from coil, flux linkage decreases and current is induced in the opposite direction to attract the magnet.
- ii. peak voltage is: $1.5 \times 5 = 7.5 \text{ mV}$

$$E = N \frac{d\phi}{dt} \Rightarrow \frac{d\phi}{dt}_{\max} = \frac{E_{\max}}{N} = \frac{7.5 \times 10^{-3}}{350} = 2.14 \times 10^{-5} \text{ Wb s}^{-1}$$

- c. Both the sign and amplitude of the induced e.m.f. observed on the c.r.o. waveform will be unchanged.
- d. higher peaks (similar amplitude + and -)
positive and negative peaks closer together
narrower / sharper peaks
sets of peaks closer together
- e. Yes, the magnetic effect of the induced e.m.f. will provide a slight resistive force to slow down the motion of the rotating axle.
The slower motion will induce an e.m.f. with peaks further apart from which the measurement of rate of revolution would be taken to be slower than it was.

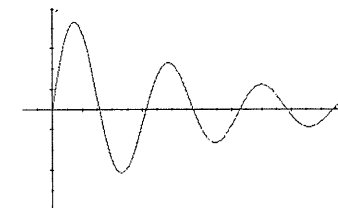
8. [SAJC/H2/Prelims 2010/P2/4]

- a. Refer to definitions in your lecture notes.
- b. i. Pointing upwards.

$$\text{ii. } F = BIL \Rightarrow B = \frac{F}{IL} = \frac{16000}{(10.0)(0.300)} = 5330 \text{ T}$$

- iii. Advantage: no wear and tear due to moving parts, low noise
Disadvantage: high magnetic field is required

- b. i. Changing direction
Decreasing amplitude

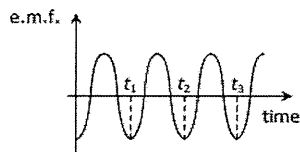


- ii. Changing direction: As the south pole approaches, the induced current produced in the coil flows in such a way that a repulsive force is produced. When the south pole withdraws from the coil, the induced current flows in the opposite direction so as to produce an attractive force.

Decreasing amplitude: The induced current flowing in the coil will generate heat in the coil. The energy lost is not recovered by the system.

9. [TJC/H2/Prelims 2010/P3/3]

- a. The magnetic flux linkage of the coil is the product of the magnetic flux through the coil and the number of turns of the coil.
- b. $X \rightarrow Y$ direction
- c. When the current increases, the magnetic flux density of the solenoid XY increases.
Flux linkage with the small coil increases.
According to Faraday's law, an e.m.f. is induced in the small coil, since there is a change in the magnetic flux linkage through the small coil.
- d. Negative cosine graph



10. [VJC/H2/Prelims 2010/P2/2]

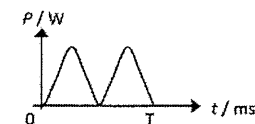
- a. Faraday's Law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
- b. i. there is no change in flux linkage of the coil, average induced e.m.f. is zero.
- ii. Time for coil to move out of the magnetic field: $\Delta t = \frac{\text{diameter of coil}}{v} = \frac{5.0 \times 10^{-2}}{15.0} = 3.33 \times 10^{-3} \text{ s}$
- Average induced e.m.f.: $E_{\text{ave}} = \frac{\Delta \Phi}{\Delta t} = \frac{NBA - 0}{\Delta t} = \frac{50 \times (2.0 \times 10^{-3}) \left[\pi \left(\frac{5.0 \times 10^{-2}}{2} \right)^2 \right]}{3.33 \times 10^{-3}} = 0.0590 \text{ V}$
- c. i. $E = BLv = (2.0 \times 10^{-3})(5.0 \times 10^{-2})(15.0) = 1.50 \times 10^{-3} \text{ V}$
- ii. N is at a higher potential.

11. [YJC/H2/Prelims 2010/P2/4]

- a. $\phi = BA = (45 \times 10^{-3}) \times \pi \times \left(\frac{0.140}{2} \right)^2 = 6.93 \times 10^{-4} \text{ Wb}$
- b. i. 1. $\Delta \Phi = N \Delta \phi = 850 \times (0 - 6.927 \times 10^{-4}) = 0.589 \text{ Wb}$
2. $E_{\text{average}} = \frac{\Delta \Phi}{\Delta t} = \frac{0.5888}{120 \times 10^{-3}} = 4.91 \text{ V}$
- ii. Zero average e.m.f. induced because the overall change in magnetic flux is zero.

12. [MI/H2/Prelims 2010/P3/6 (part)]

- a. i. An e.m.f. is induced in coil Y.
- ii. Faraday's law states that the induced e.m.f. is proportional to the rate of change of magnetic flux linkage.
- b. $V_Y I_Y = V_X I_X \Rightarrow I_X = \frac{V_Y I_Y}{V_X} = \frac{(12) \times (3.0)}{240} = 0.15 \text{ A}$
- c. 2 power cycles for a cycle of voltage
Positive sinusoidal variation with zero to max power



13. [MJC/H2/Prelims 2010/P2/3]

- a. $\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{8}{3}$
- $V_{\text{power station}} = V_{\text{home}} \left(\frac{8}{3} \right)^n$ where n is the number of transformers required
- $\frac{230000 \text{ V}}{240 \text{ V}} = \left(\frac{8}{3} \right)^n$
- $n = \log_{\frac{8}{3}} \left(\frac{230000 \text{ V}}{240 \text{ V}} \right) = 6.999 = 7 \text{ (nearest integer)}$

Assumption: any one

- The transformers are ideal with 100% efficiency.
 - The resistance of the primary and secondary coils r_p and r_s are zero, so that no energy is lost in the coils.
 - There are no magnetic flux losses and hence both coils have the same flux through them.
- b. The purpose of the iron core is to confine the magnetic field lines to ensure maximum magnetic flux linkage between the primary and secondary coils.
- c. $P_{\text{ave}} = \frac{V_{\text{r.m.s.}}^2}{R} = \frac{(240)^2}{58} = 993 \text{ W}$
- d. Hair dryer / Vacuum cleaner / Microwave oven / any appliance that has about 1000 W power rating